
Observed video family
$\{(s_j, x_j)\}_{j=1}^{L}$
s_1 s_2 s_3 ⋯ s_L
VAE Encoder E_ϕ
Latent set
$\{(s_j, z_j)\}_{j=1}^{L}$
z_1
z_2
z_3
⋯
z_L
Family Generator F_θ
Family code $\boldsymbol{b}$
Query at observed control values s_j
Query Decoder R_ψ
$\hat{z}_j$
$\mathcal{L}_{rec} + \mathcal{L}_{fm}$
Reconstruction supervision (at observed s_j)